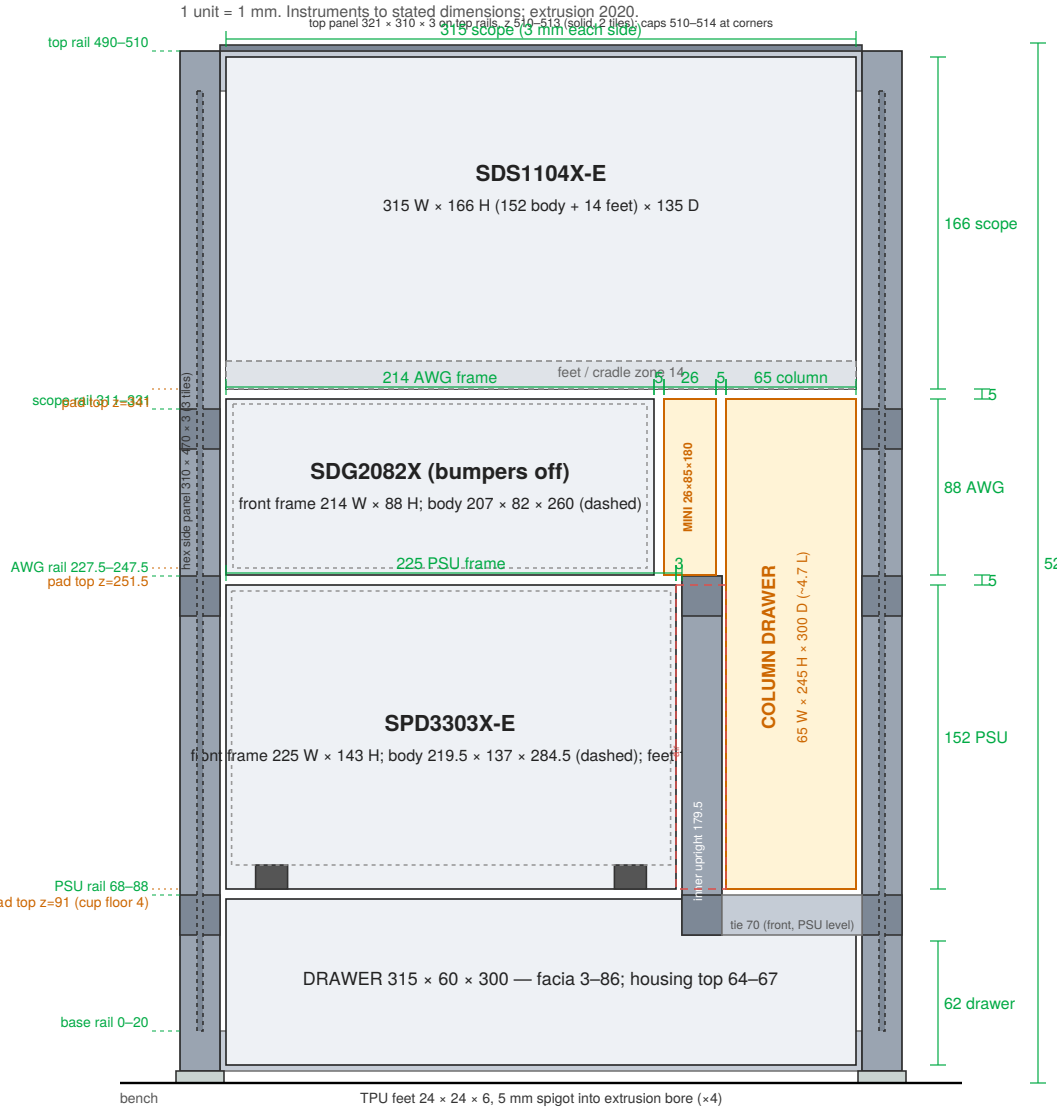
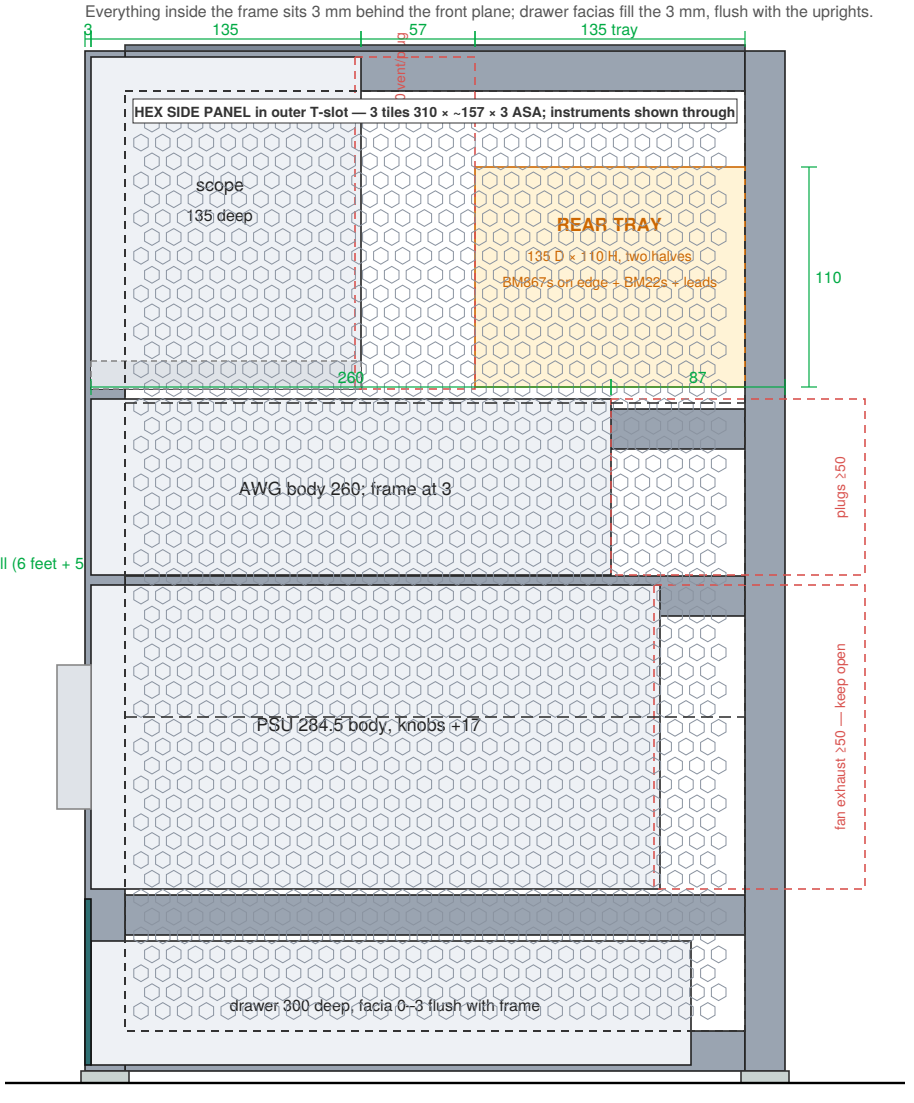
FRONT ELEVATION

1 unit = 1 mm. Instruments to stated dimensions; extrusion 2020.



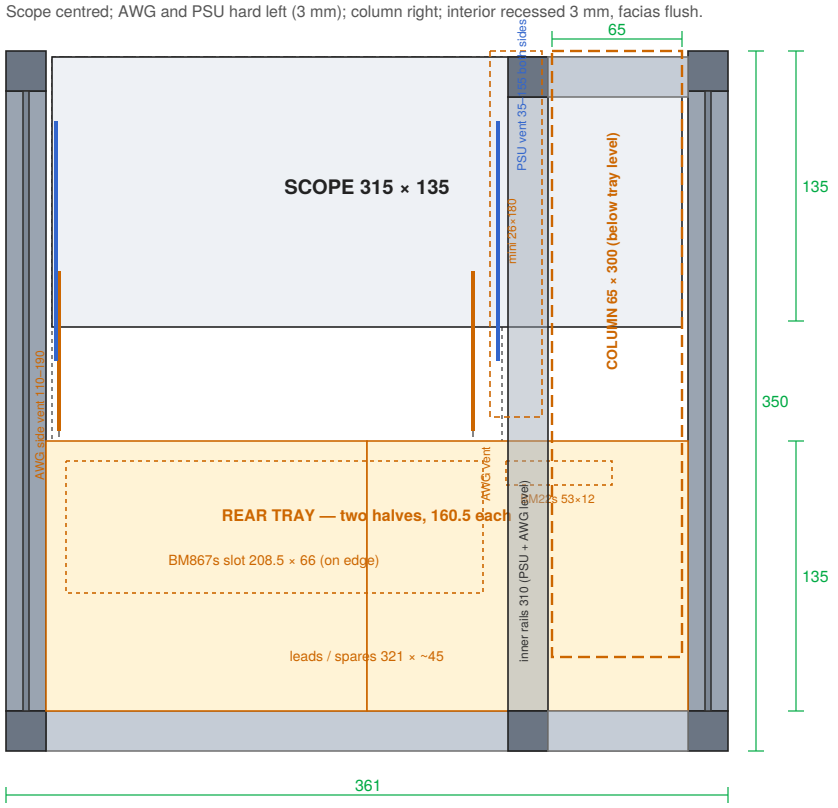
SIDE ELEVATION (right side, front at left)

Everything inside the frame sits 3 mm behind the front plane; drawer facias fill the 3 mm, flush with the uprights.



PLAN (front at top)

Scope centred; AWG and PSU hard left (3 mm); column right; interior recessed 3 mm, facias flush.



2020 EXTRUSION CUT LIST

Uprights	4	510
Side rails (front–rear), 5 levels	10	310
Rear cross rails (base + top)	2	321
Inner rails (PSU + AWG level)	4	307
Inner uprights	2	179.5
Ties inner→outer (PSU front+rear, AWG rear)	3	70
Total		7,579 mm → buy 8 × 1,000 mm

Joints: 40 × 2020 corner brackets, 80 × M5 T-nuts, 80 × M5×8 button head. Feet: 4 printed.

LEVELS (z from bench, mm)

Base rails	0–20	drawer runs on inside faces; drawer body 3–65
PSU rails	68–88	4 foot cups, floor 4 above rail = z 91; feet 11.5, body 102.5–239.5, frame 99.5–242.5
AWG rails	227.5–247.5	4 corner pads 3.5 behind the front frame, top z=251.5; frame 248–336
Scope rails	311–331	SLED: 2 × 250 T-slot sliders 15 tall + front bar; cups/pads top z=341; scope top 507; pulls 140
Top rails	490–510	brace only; tray hangs off rear uprights below
Inner sub-frame	68–245	x 251–271, y 3–350; front inner upright and front tie at y 3–23; column 273–338

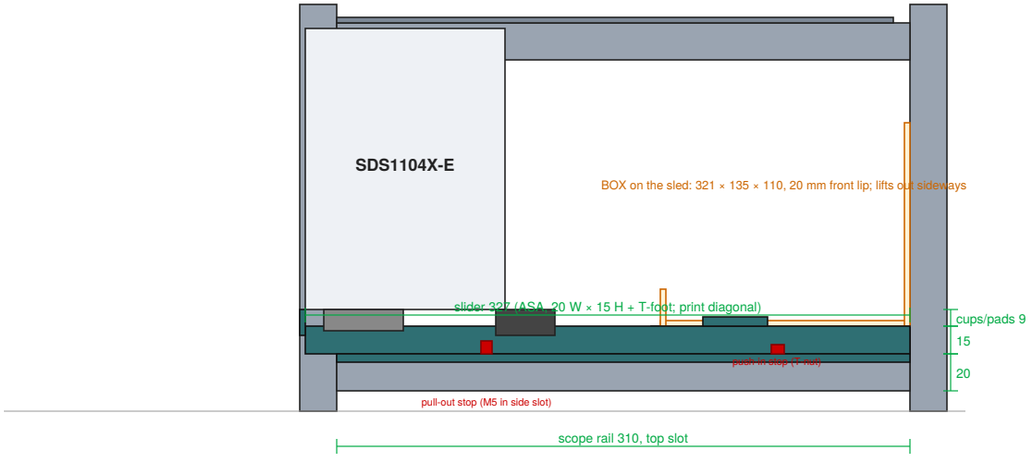
PRINTED PARTS (all under 256 mm)

- 8 × cradle corners (PSU/AWG: 4 left outer + 4 right inner); scope sled: 2 × slider 327 + front bar + 2 rear bars (page 2)
- 2 × rear box halves 160.5 × 135 × 110 joined (dovetail), 20 mm front lip, rides the sled — access: page 2
- 2 × drawer halves (front/back) 315 × 60 × 300, joined; facia 315 × 83 × 3; 2 × runners; housing top 315 × 300 × 3 (2 tiles)
- 1 × column drawer 65 × 240 × 300 + facia; 1 × mini slot 26 × 83 × 180; shared housing top 96 × 300 × 3 at 333–336
- Side tiles 3 mm ASA, 4 mm solid border, clip into rail slots, one per band between rails: 20–68, 88–227.5, 247.5–311, 347–490 (band 331–346 open
- 4 × TPU feet; 2 × TPU sled-stop caps; 4 × ASA caps; 2 × door hinge knuckles + 2 × magnet clips (4 × Ø10×2 magnets); 1 × probe hanger

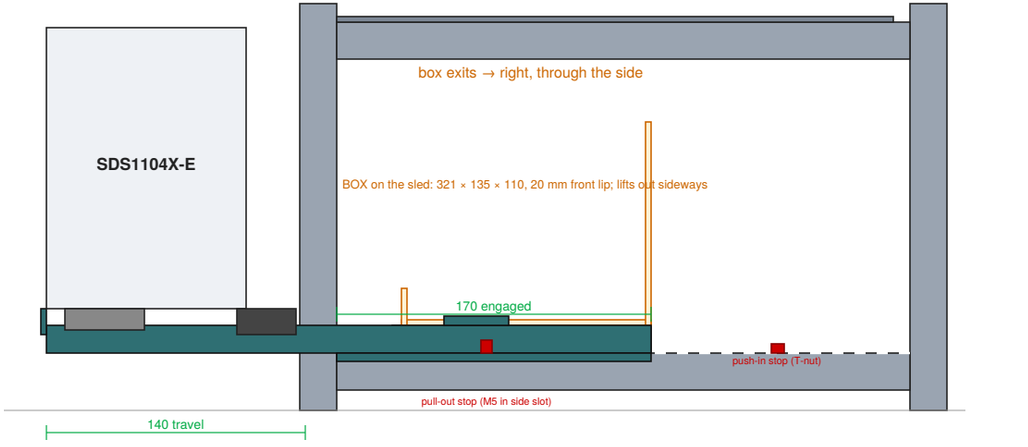
RULES BAKED IN

- Width 361: 315 scope + 3 mm each side + 2 × 20 uprights. PSU/AWG hard left; screens align left
- Height 520 incl. feet and caps (limit 530): 5 mm reveals throughout; scope flat = 166, 3 mm under the top panel
- PSU vents 35–155 both sides: inner posts sit at y 0–20 and 330–350, clear of the vent; channel open end to end
- Interior 3 mm behind the frame front; facias 0–3 flush; all reveals 5 mm (3 mm to the uprights, set by the scope); rear open
- AWG frame 3.5 proud each side, 3 top/bottom; PSU frame 2.75 each side, 3 top/bottom, 3.5 deep — cradle pads and cups sit behind both frames
- Scope cradle equalises front/rear foot heights so it sits level

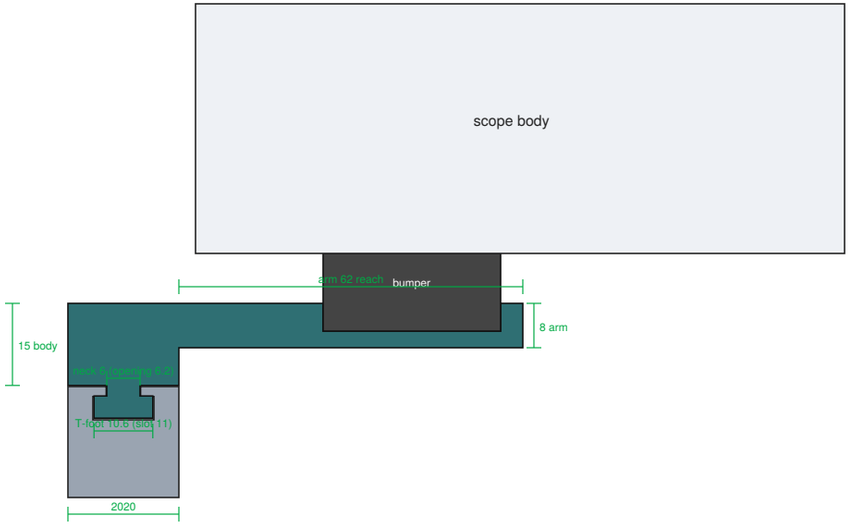
A — SCOPE SLED, closed (side view, front at left)



B — SLED pulled 140: box now at y 55–190, in the clear zone; remove right top tile, slide box out to the right (350 clear)



C — SECTION through scope rail and slider (×3)



LOADS AND CHECKS

Scope 3 kg → 15 N per slider. Closed: pure compression on the rail top, nothing to check.
Pulled 140: scope CoG sits ~90 mm ahead of the rail's front end with 170 mm of slider still in the slot; box (~1 kg) rides behind the scope.
Lever ≈ 0.5 : the rear of the T-foot pulls UP on the aluminium slot lip with ~8 N; the front bears DOWN on the rail end with ~30 N.
Slot lip: aluminium, 1.8 mm — this joint carries 3D-printer gantries; 16 N is nothing.
T-foot / neck (ASA): 16 N over 90 mm of engaged foot → < 0.1 MPa bearing, ~0.5 MPa shear at the neck. Yield 40 MPa.
Slider body 20 × 15 ASA as a 140 mm cantilever with 15 N at the tip: bending stress ≈ 2.8 MPa; sag ≈ 0.9 mm. Acceptable.
(An 8 mm body would sag ~5 mm — hence 15 mm, and the scope rails drop 7 mm to keep the scope at the same height.)
Play: 0.2 mm slot clearance × lever ≈ 0.5 mm droop at the bezel when fully out. Print the T-foot 10.6 wide for a light sliding fit; sand if tight.
Arms 62 mm × 8 mm thick carrying the bumper cups: same section as the PSU cradle, 15 N → ~1 MPa.
Front bar 315 × 14 × 12 with J-pull = handle + reveal filler; two rear cross bars 321 × 20 × 15 carry the box. Sled = 2 sliders + 3 bars, one rigid frame.
Box guides on the sled: 10 mm lips along the front and rear bars (full width, 1 mm play) and an end stop on the left slider — the box slides in from the right between the lips until it meets the stop.
ACCESS: pull sled 140 → box sits at y 55–190 → fold down the FRONT section (y 20–250) of the top-right side tile (bottom hinge) → slide box out to the right (321 long, 350 clear). Scope never lifts.
Stops: push-in = T-nut in the top slot behind the slider; pull-out = M5 + T-nut in the rail's inner side slot that the rear arm meets at 140, with a TPU cap on the screw head so the sled stops softly.
Lube: PTFE dry spray in the slot. Leads: 150 mm slack on mains and probes behind the scope.
DOOR: 230 × 143 × 3 ASA hex tile, bottom-hinged on two printed knuckles (3 mm pin), axis z 341 outboard of the upright, opens 180. Catch: 2 pairs of Ø10 × 2 magnets at the top corners — door side
Layout change for v13: scope rails 311–331 (were 318–338). Scope, tray, top panel, everything else unchanged.